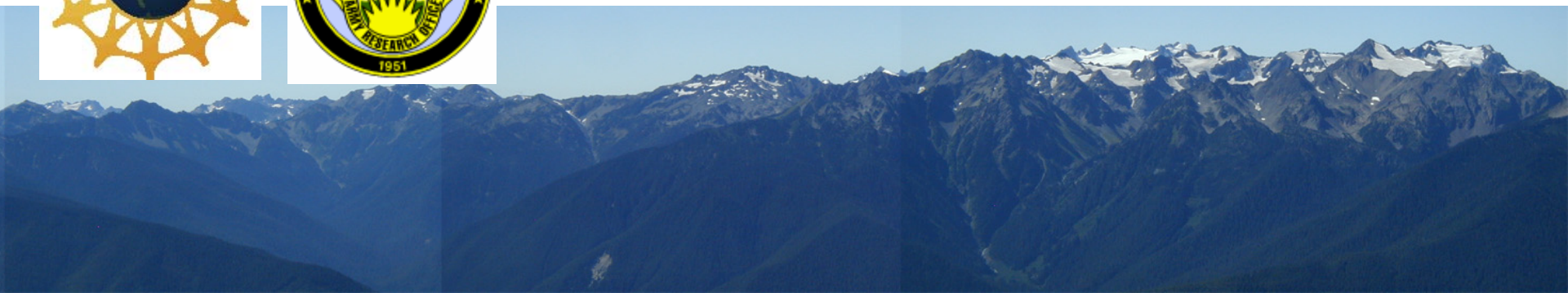
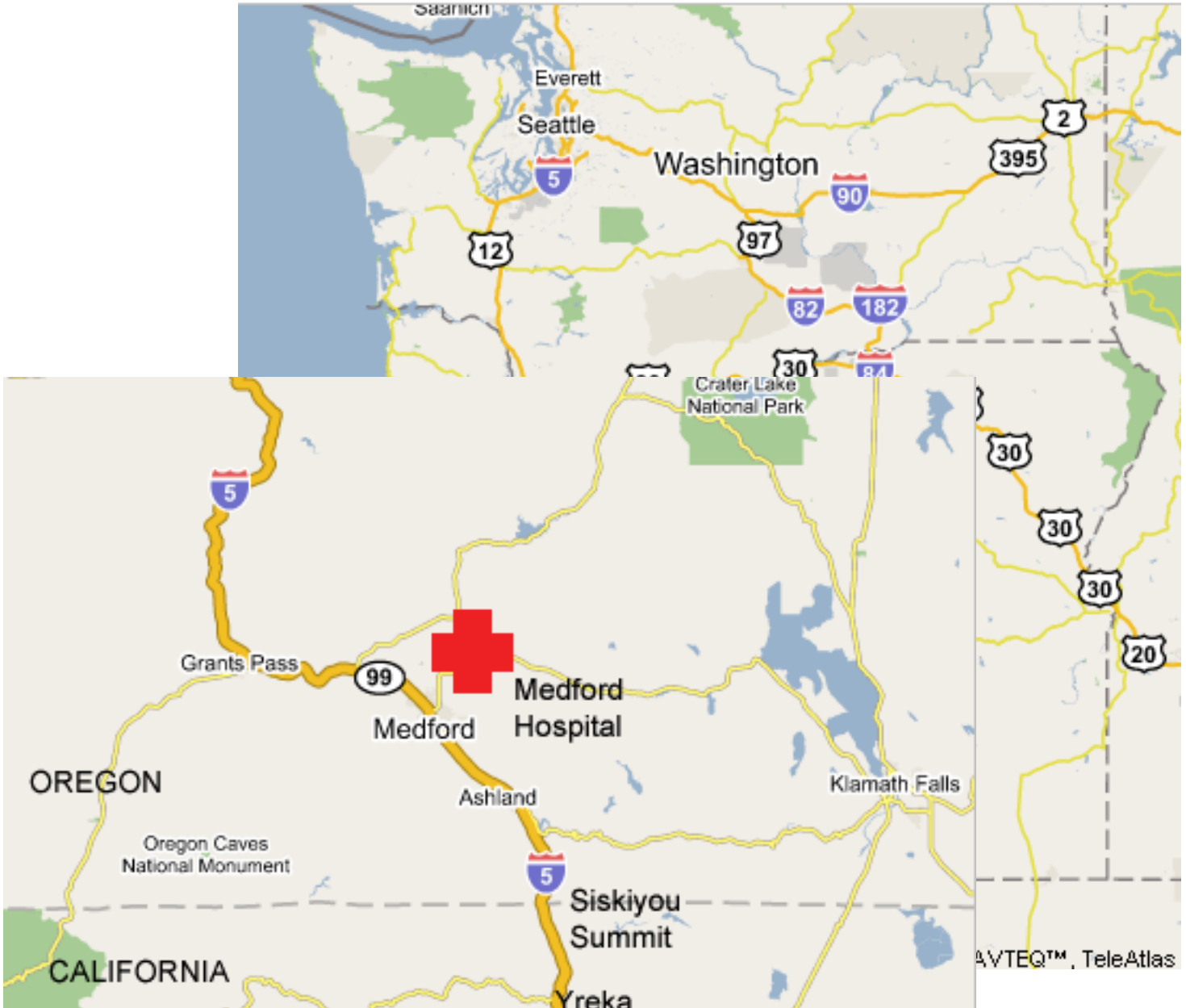
# When Physicists Build Quantum Algorithms

**Dave Bacon**
Department of Computer Science & Engineering
University of Washington

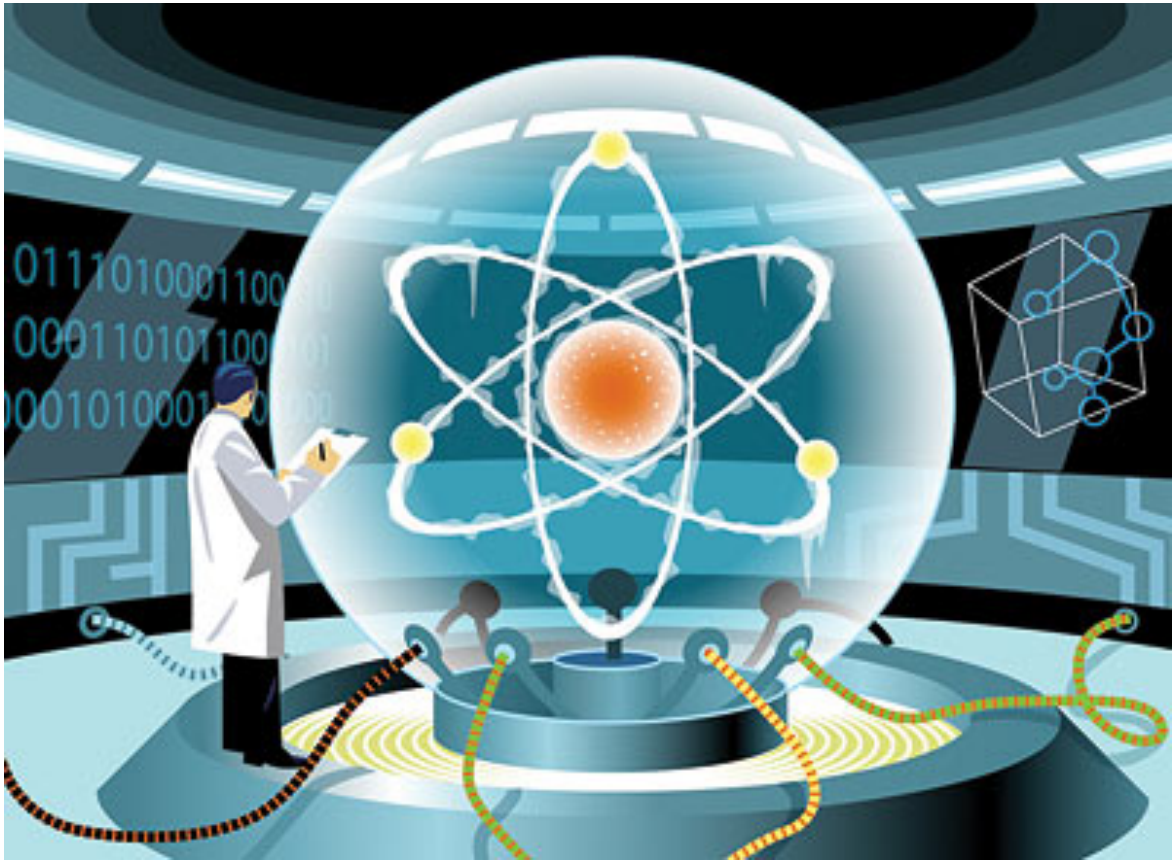
*funded by*



# All Apologies



# Quantum Computers



Quantum Computing for Science’s sake

# Traditional Excitement

1. Building a quantum computer

2. Quantum computers break cryptosystems

3. Quantum cryptography as physics based security

# Emerging Themes

**[REVOLUTION]** Quantum computing as key to progress in theoretical computer science

**[STAND ON HEAD]** Understanding power of quantum computers ⇒ new efficient classical algorithms for simulating quantum systems

**[PHYSICISTS!]** Tools and methods developed by physics used for useful computer science (new quantum algorithms)

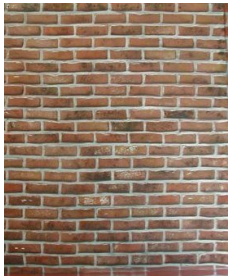
# Quantum Computer?



"How can we simulate quantum mechanics? ...Can we do it with a new kind of computer …a quantum computer? It's not a Turing machine but a machine of a different kind."

- Richard Feynman (1981)



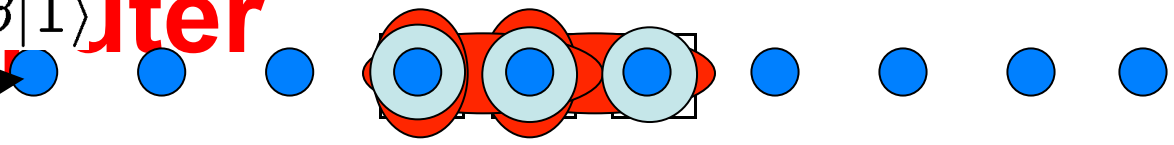


simulating quantum theory

# The Ideal Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

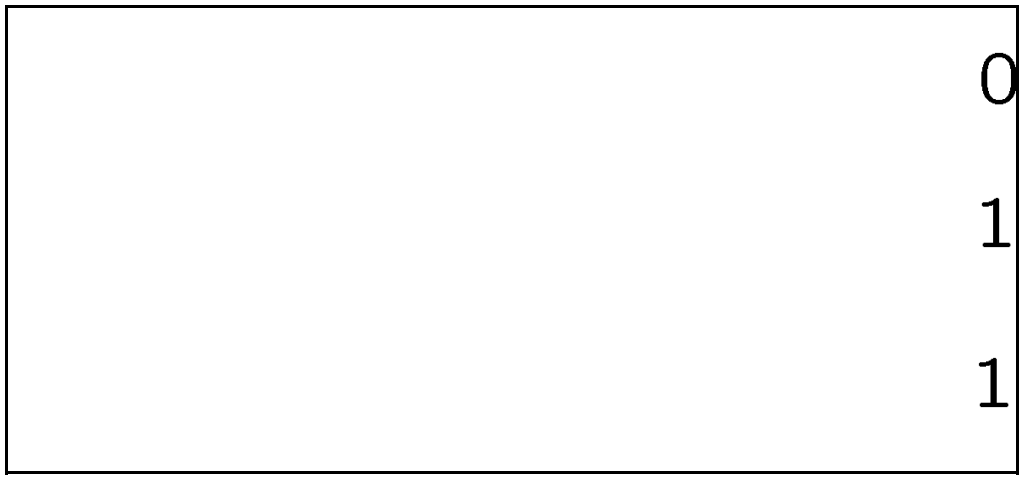
qubits




$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

Quantum algorithm

1. Initialize
2. Series of unitary gates
3. Measurement



# Major Triumph I: Factoring

FACTOR an *n* digit number

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Classical running time: $O(e^{n^{\frac{1}{3}}})$

Quantum running time: $O(n^2 \log n)$ [Peter Shor 94]



Public Key Cryptosystems

RSA

S[illegible]man

[illegible]tic

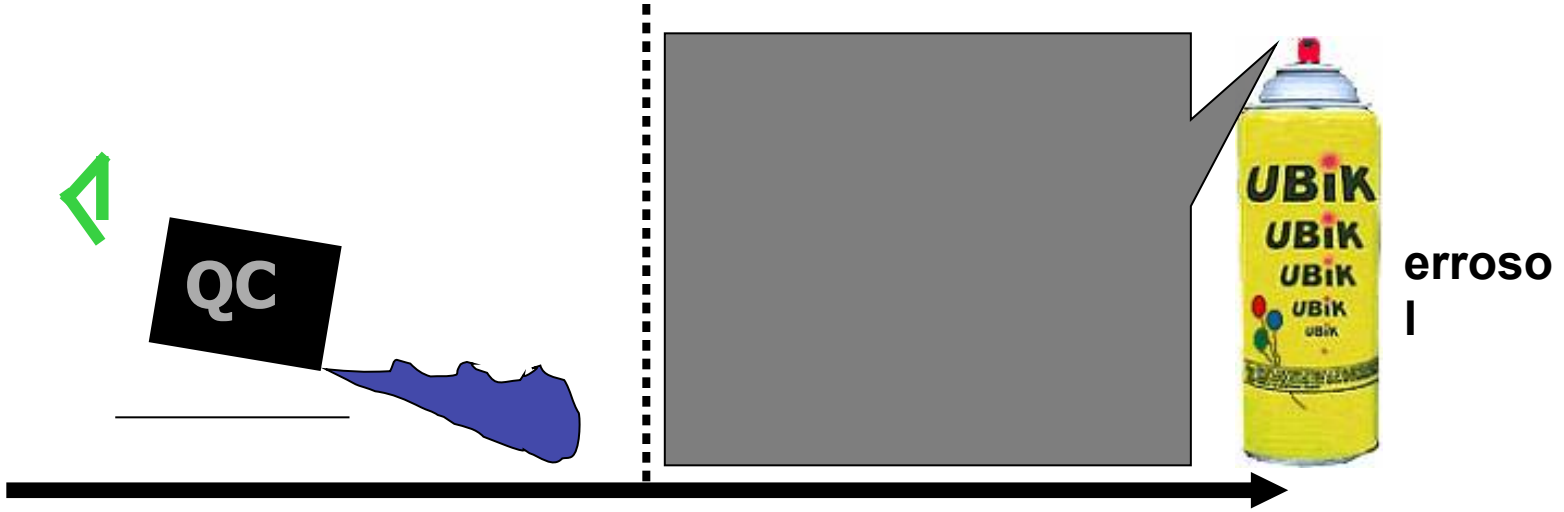
Curves

**BROKEN**



Also : Grover's polynomial quantum speedup

# Major Triumph II: Quantum Error Correction




**Threshold “Theorem” for Fault-Tolerant Quantum Computation**

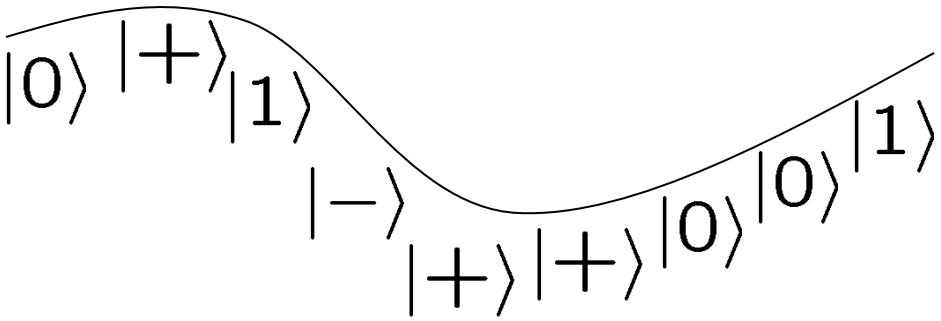
[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev, Gottesman, Preskill, Shor, + others]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

# Major Triumph III: Quantum Cryptography



ALICE






BOB

**Quantum key distribution:**

Distribute shared secret keys while detecting eavesdropper

Commercial products:

MagiQ

id Quantique

# Major Triumph IV: Quantum Experiments

New levels of coherent control of quantum systems

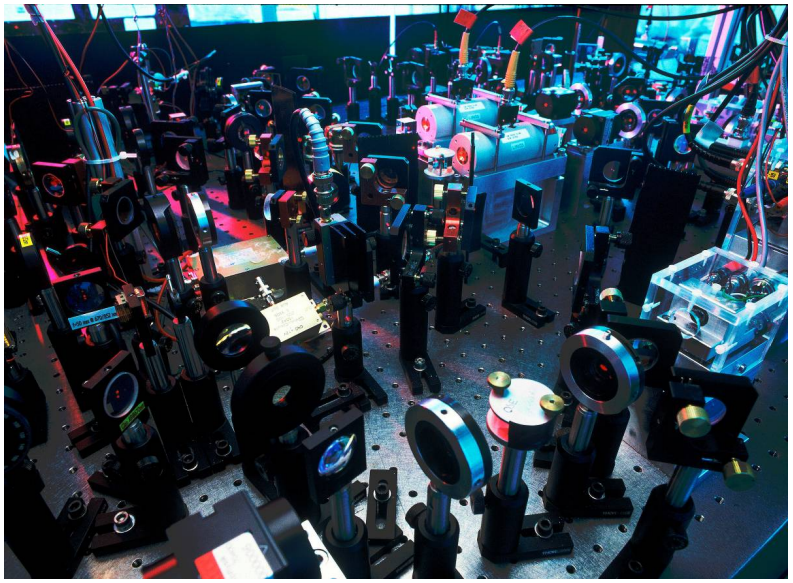


Photo: C. Lackner

8 qubit entangled state in ion traps (Innsbruck)

[*Nature* **438**, 643-646 (2005)]

$$\frac{1}{2\sqrt{2}}(|10000000\rangle+|01000000\rangle+|00100000\rangle+|00010000\rangle+|00001000\rangle+|00000100\rangle+|00000010\rangle+|00000001\rangle)$$

"w state"

# Quantum Computing Roadmap

**Build large scale quantum computer**

**Factor numbers**

**Make NSA happy**

Quantum Computing for Science's sake?

# Emerging Themes

**[REVOLUTION]** Quantum computing as key to progress in theoretical computer science

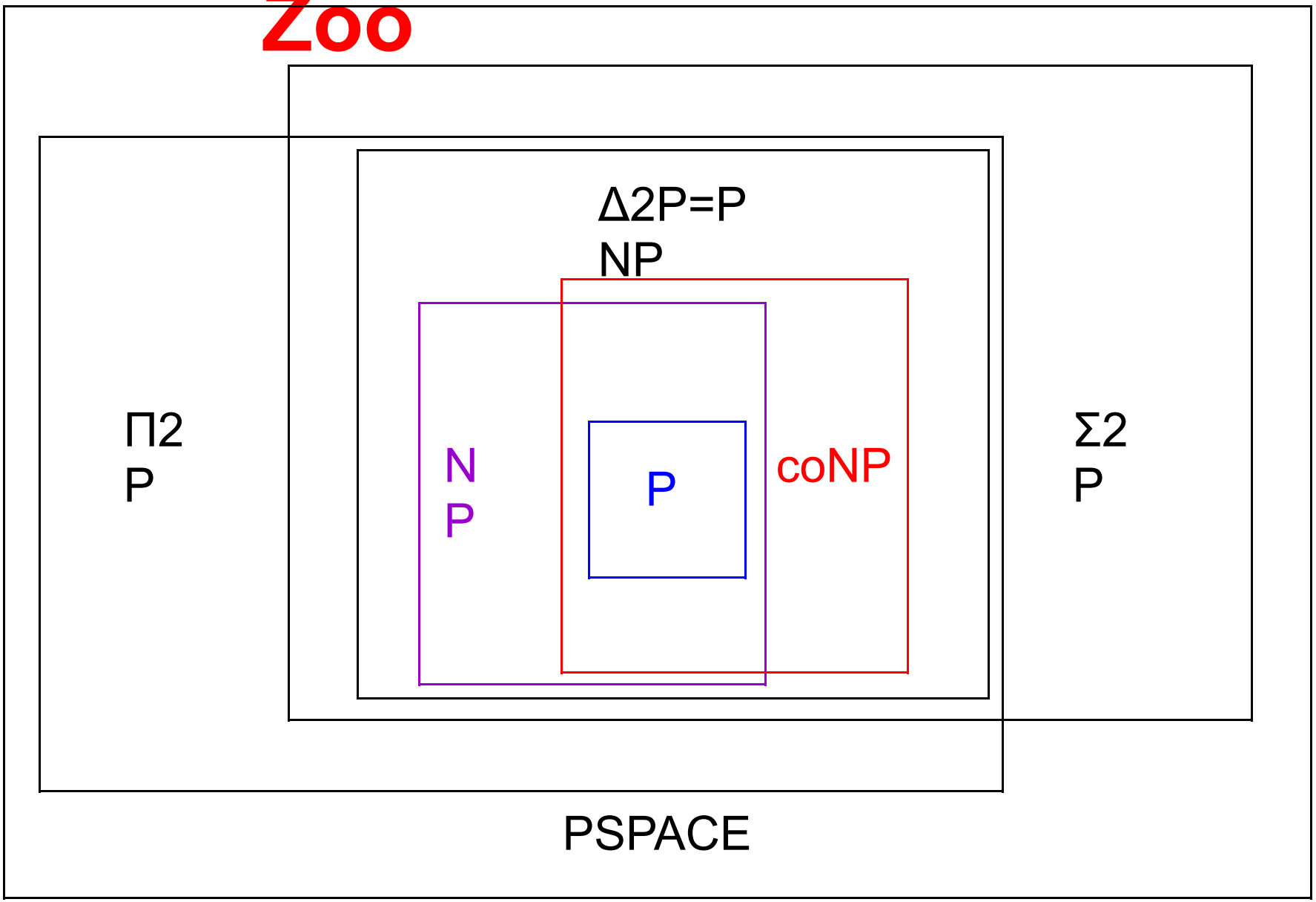
**[THE PHYSICISTS ARE COMING!]** Tools and methods developed by physics used for useful computer science (new quantum algorithms)

**[STAND ON HEAD]** Understanding power of quantum computers ⇒ new efficient classical algorithms for simulating quantum systems

# Revolution

The universe is quantum, so should we be surprised that the theory of quantum computation is more “fundamental” and hence more useful?

# The Complexity Zoo



# Example

Boolean function *f: n* bits to *1* bit

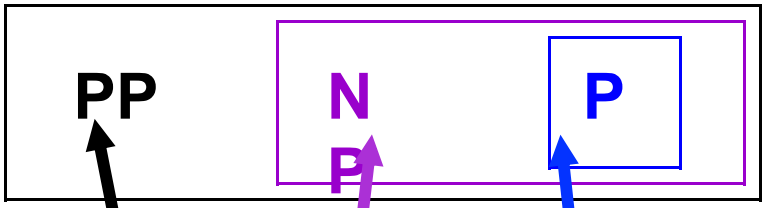
$$f(x_1,\ldots,x_n) = (x_1 \vee \bar{x}_4 \vee x_5) \wedge (x_1 \vee \bar{x}_2 \vee x_7) \wedge \cdots \wedge (\bar{x}_4 \vee x_5 \vee x_6)$$

OR AND NOT clause

*K* variables per clause

*M* clauses

Problem size = O(*KM*)




**P**: Polynomial

**NP**: Nondeterministic Polynomial

**PP**: "Probabilistic Polynomial"

2SAT (K=2) Does *f(x)=1* have at least one solution?

3SAT (K=3) Does *f(x)=1* have at least one solution?

MAJSAT: Does *f(x)=1* have more than *2n-1* solutions?

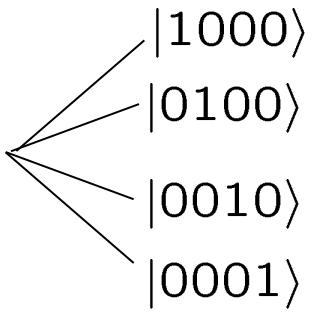
# From Philosophy to CS

"Many-worlds" and "anthropic reasoning":



Aaronson

$$\frac{1}{2\sqrt{2}}(|1000\rangle + |0100\rangle + |0010\rangle + |0001\rangle)$$




Postselection:
Select world where last bit is one.

**PostBQP**: Polynomial time quantum algorithm followed by postselection.

How powerful is such a quantum computer?

**PostBQP=PP** [Aaronson 2005]

# Revolution

The universe is quantum, so should we be surprised that the theory of quantum computation is more "fundamental" and hence more useful?

**PP** closed under intersection?

open problem (over 14 years)

solved in [Beigel, Reingold, and Spielman, 1995]

complex proof using logical AND of two majority functions by the sign of a low-degree rational function.

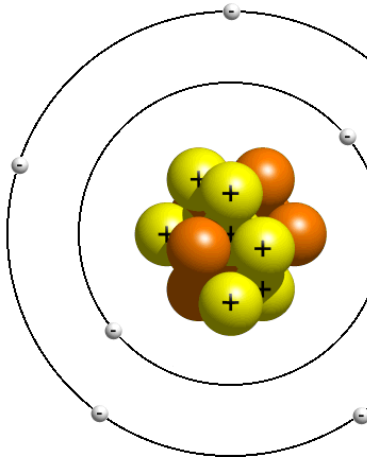
**PostBQP** is trivially closed under intersection.

Scott Aaronson's proof "explains" the prior proof using the quantum theory of computation!

# When Physics and Computer Science Collide

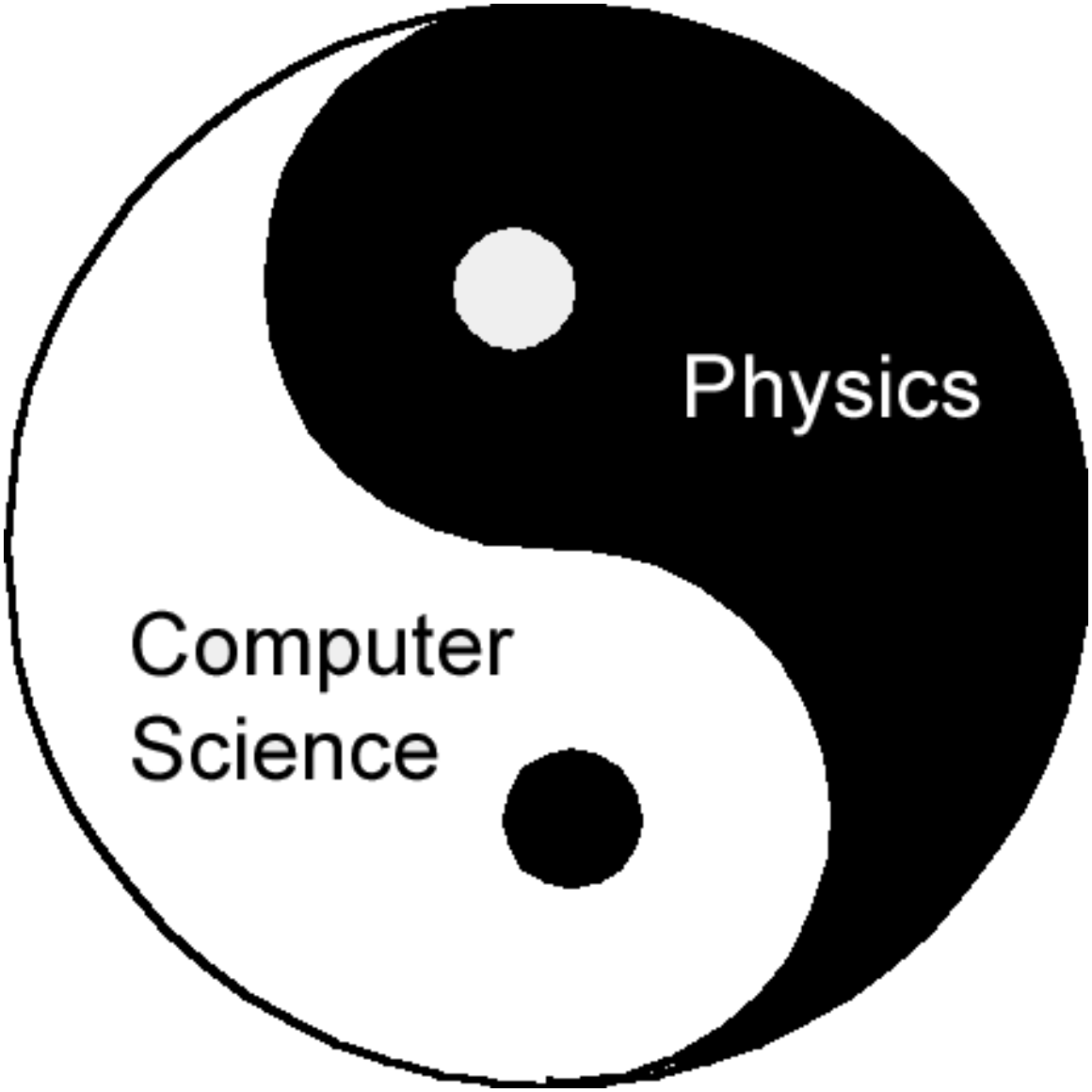
**Email I received from Wim van Dam about FOCS 2006:**

Leslie Valiant in his great talk on 'accidental algorithms': "We have all these complexity classes, and we don't know at all if they are different. It would be a nightmare if they turn out to and so on." <comment fro lly be so bad> "Yes it possible nightmare wo hysicist* who proved i










But what can you, as a physicists, contribute to this dialogue?

# Emerging Themes

**[REVOLUTION]** Quantum computing as key to progress in theoretical computer science

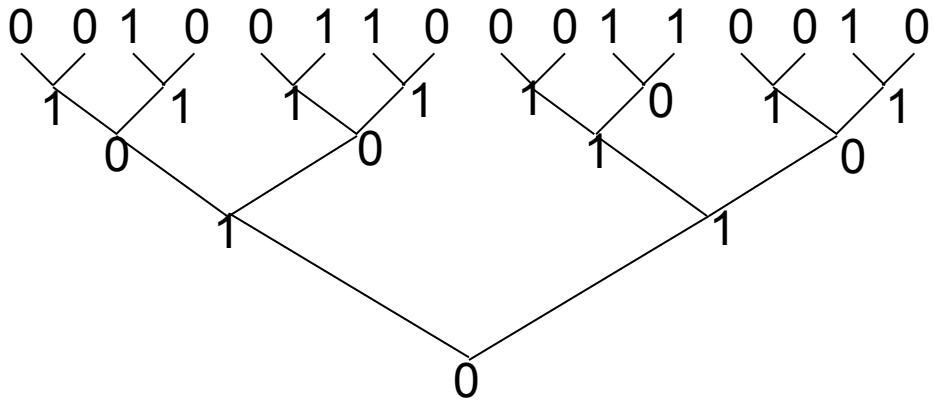
**[THE PHYSICISTS ARE COMING!]** Tools and methods developed by physics used for useful computer science (new quantum algorithms)

**[STAND ON HEAD]** Understanding power of quantum computers ⇒ new efficient classical algorithms for simulating quantum systems

# NAND Tree Problem

NAND

| XY | (XY) |
|---|---|
| 00 | 1 |
| 01 | 1 |
| (1) | 1 |
| (2) | 0 |




How many nodes (of *N*) do you need to query to determine root?

Classically: $N^{0.753...}$ $0.753\ldots = \log_2(1+\sqrt{33}) - 2$

Quantum:
?

# NAND Tree Problem

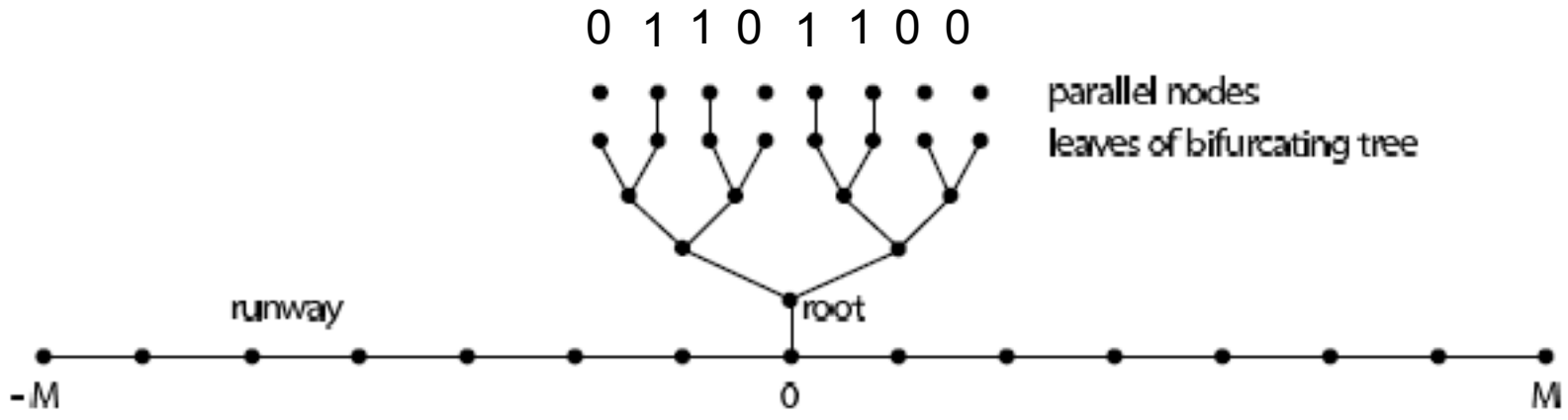



Figure 2: The full Hamiltonian $H_O + H_D$

WAH?

Fig. from quant-ph/0702144 Farhi, Goldstone, Guttman

Hamiltonian = Discrete Laplacian on above graph

Right moving wave packet on left runway.

Passes through? NAND Tree = 0

Scatters back? NAND Tree=1

Running times?

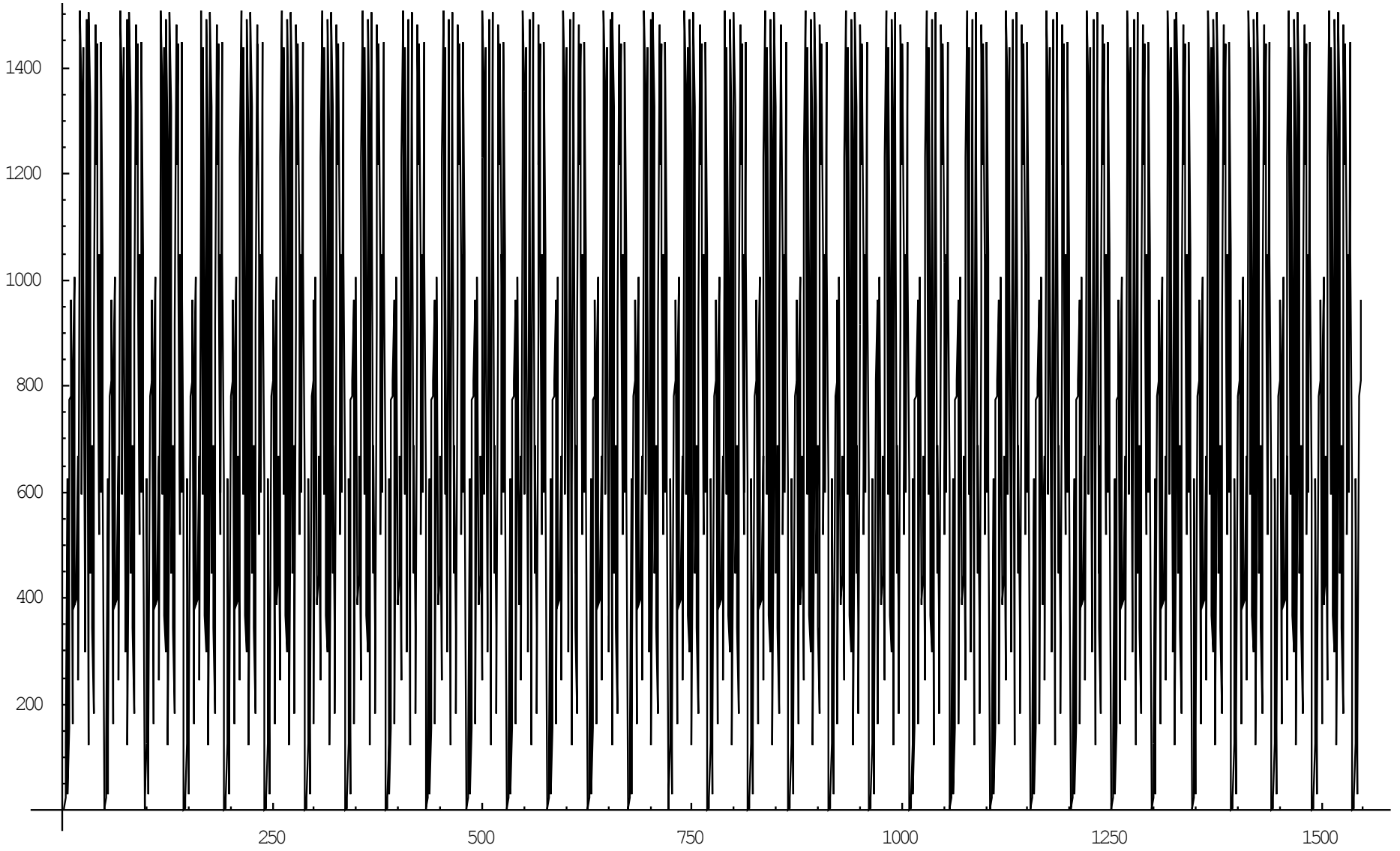
N1/2

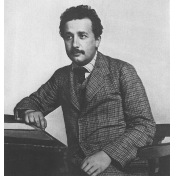
Quantum NAND tree algorithm: N1/2 time

# Shor's Algorithm



N=1547, r=5, Period = 48.  48+1=49  GCD(1547,49)=7





# Period Finding

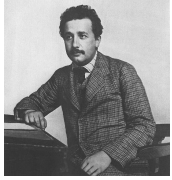
$$f(s) = r^s \bmod N$$



$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |s\rangle \otimes |f(s)\rangle$$

But just computing everything “in parallel (universes)” doesn’t help us find the period, does it?

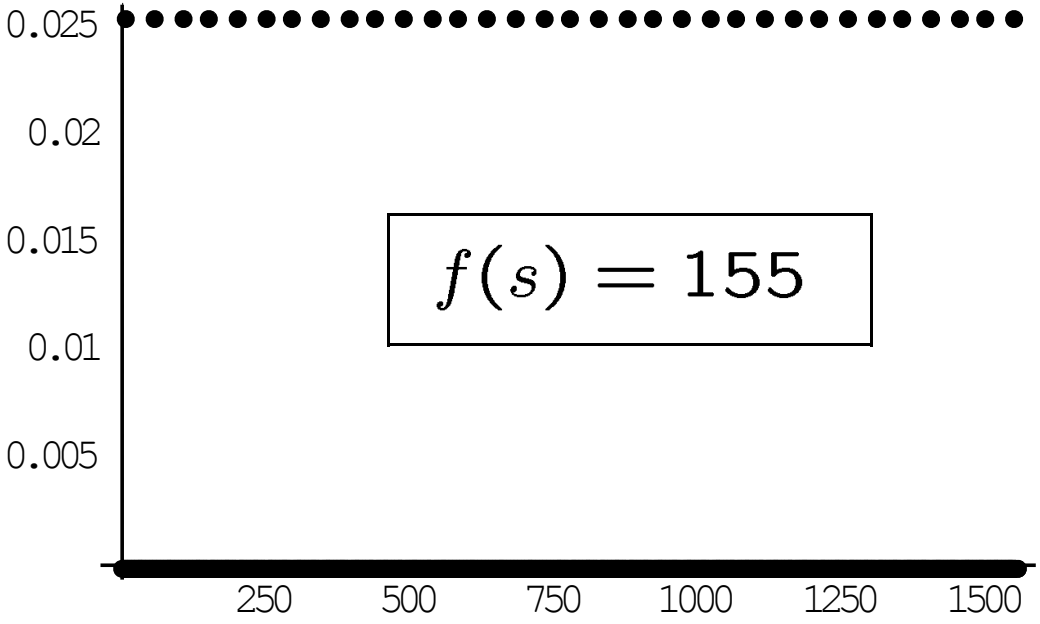
# Period Finding



$$f(s) = r^s \bmod N$$



$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |s\rangle \otimes |f(s)\rangle$$




$$f(s) = 155$$

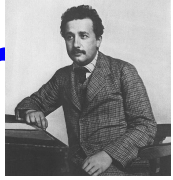
Measure second register

First register is (nearly) symmetric under translation

$$T|x\rangle = |x+1\rangle$$

$$T\rho T^\dagger \approx \rho$$

# The First Basis Change We Learn In Quantum Mechanics

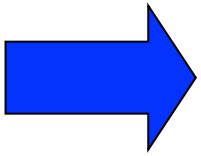


$$\frac{1}{\sqrt{N}} \sum_{s=0}^{N-1} |s\rangle \otimes |f(s)\rangle$$

symmetry under translation

$$T|x\rangle = |x+1\rangle$$

$$T\rho T^{\dagger} = \rho$$



change to momentum basis!

$$|p\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega_N^{ixp} |x\rangle$$

$$\omega_N = \exp\left(\frac{2\pi i}{N}\right)$$

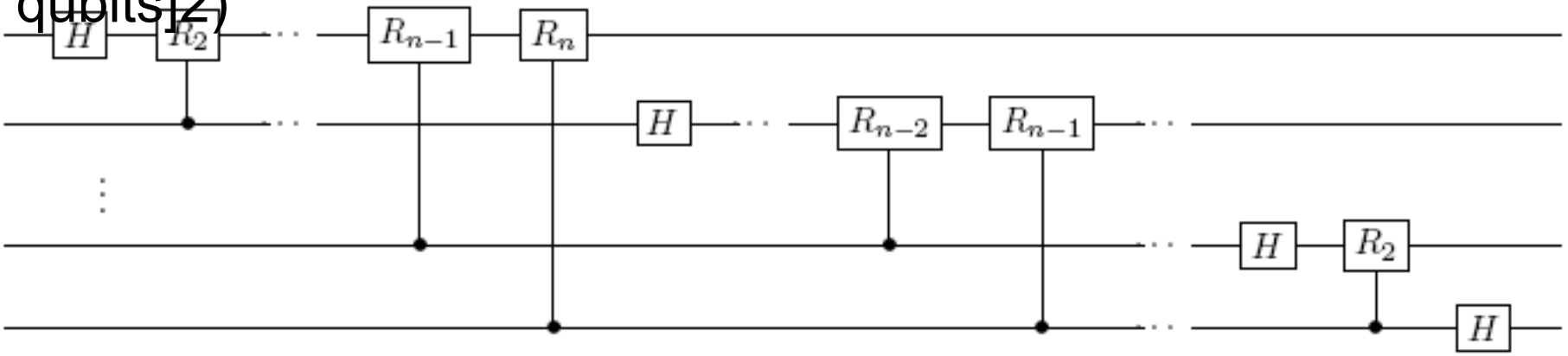
Perform a Fourier Transform!

# Quantum Fourier Transform



Efficient quantum circuit for quantum Fourier transform:

[Efficient = O([# qubits]2)




Factoring



Period Finding

Symmetry





Change Basis

Efficient Circuit

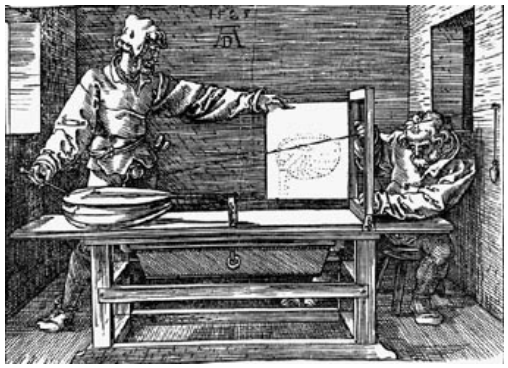




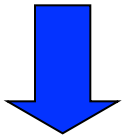
Shor's Algorithm



# Moral



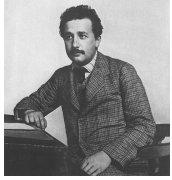
Symmetries



Quantum algorithms

Think like a physicist!

## The Second Basis Change We Learn In Quantum Mechanics?



# The Second Basis Change We Learn In Quantum Mechanics:

## Clebsch-Gordan Transform

**Singlet:**

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

**Triplet:**

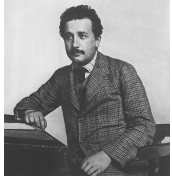
$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

**Why do we use this transform in physics?**

# Collective Unitaries



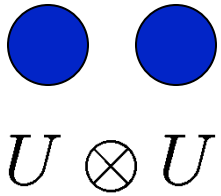
$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

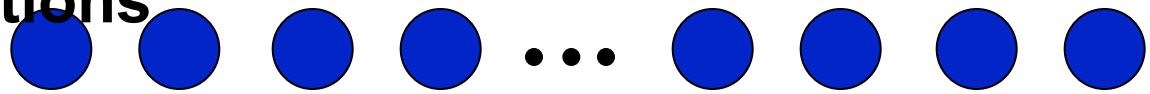
$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

$$\left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & U \end{array}\right]$$



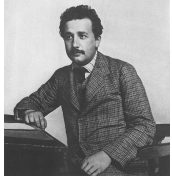
$U \otimes U$

**Symmetry Under n Collective Unitary Rotations**



$U^{\otimes n}$

# The Symmetric Group



**Transform Under SWAP**

**Singlet:**

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) \qquad -1$$

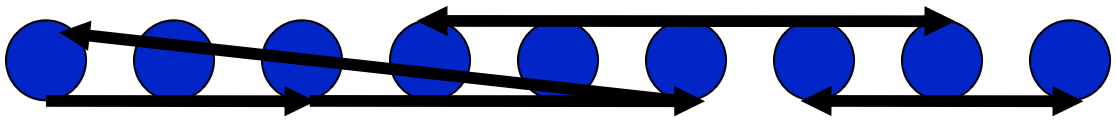
**Triplet:**

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

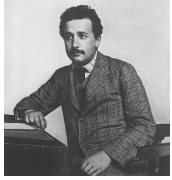
$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle) \qquad +1$$

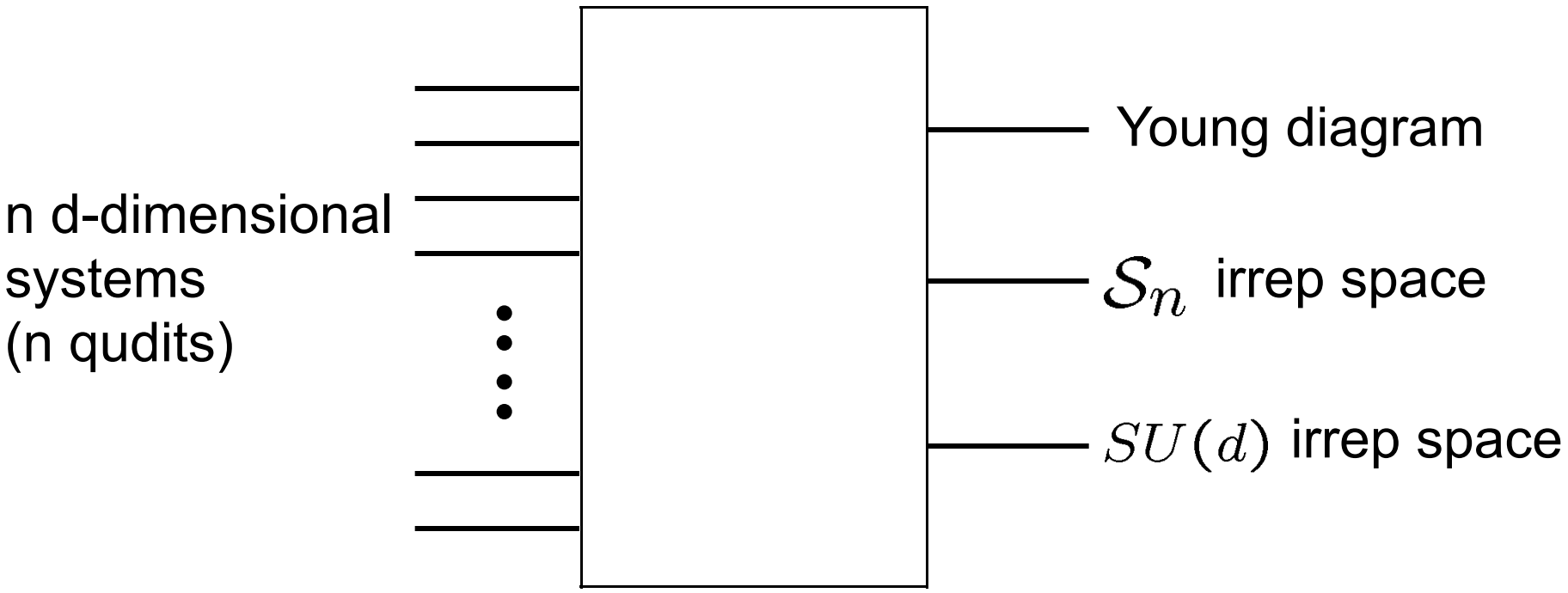
$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

**Symmetry Under Permutation of n Subsystems**



# Quantum Schur Transform

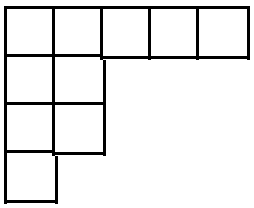





Schur duality

$$[\mathbb{C}^d]^{\otimes n} \overbrace{\cong}^{\text{Schur}} \bigoplus_{\mathcal{Y}} S_{\mathcal{Y}} \otimes U_{\mathcal{Y}}$$

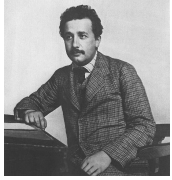
Young diagram

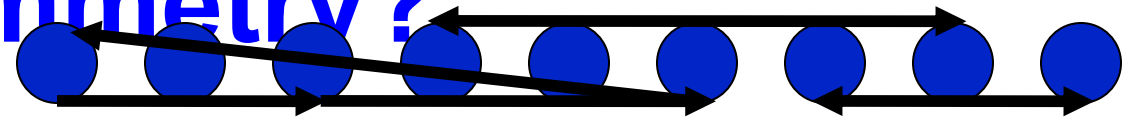


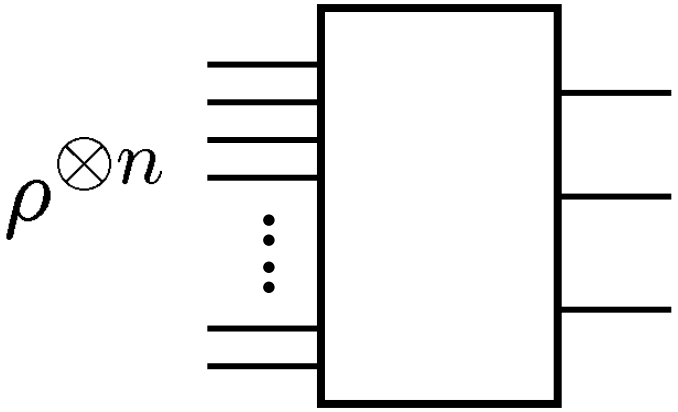
$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

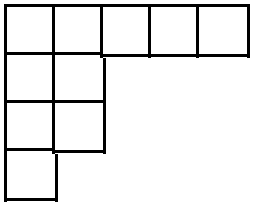
# What Use, This Symmetry?








Young diagram



$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

$$\rho = \sum_{i=1}^{d} p_i |\phi_i\rangle\langle\phi_i|$$

Spectrum

$$\left(\frac{k_1}{n}, \frac{k_2}{n}, \ldots, \frac{k_d}{n}\right)$$

"Optimal" estimate of spectrum

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

# The Plethora of Uses of Schur



Optimal spectrum estimation

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

Universal distortion-free entanglement concentration

[M. Hayashi and K. Matsumoto. Information Theory, 2004]

Universal compression with optimal overflow exponent

[M. Hayashi and K. Matsumoto. Phys. Rev. A, 2002]

Optimal quantum hypothesis testing

[M. Hayashi, J. Phys. A, 2002]

Communicating without a shared reference frame

[S. Bartlett, T. Rudolph, and R. Spekkens, Phys. Rev. Lett., 2003]

Encoding into decoherence-free (noiseless) subsystems

[J. Kempe, D. Bacon, D. A. Lidar, and K. B. Whaley, Phys. Rev. A, 2001]

Simulating lattice gauge theories on a quantum computer

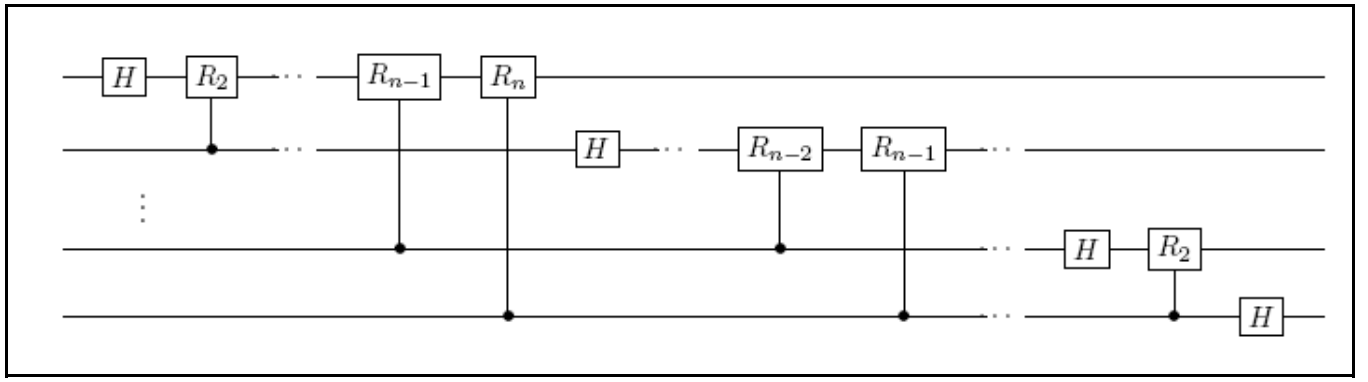
[T. Byrnes and Y. Yoshihisa, Phys. Rev. A 2006]

….

# But Efficient?



QFT




Schu
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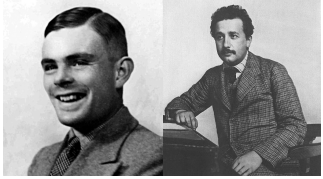
?

Who would need to efficiently calculate Clebsch-Gordan coefficients for SU(d)?



[L. C. Biedenharn and J. D. Louck, Commun. Math. Phys. 1968]

# Efficient Schur



[D. Bacon, I.L. Chuang, A. Harrow, Phys. Rev. Lett. 2006]
[D. Bacon, I.L. Chuang, A. Harrow, SODA 2007]

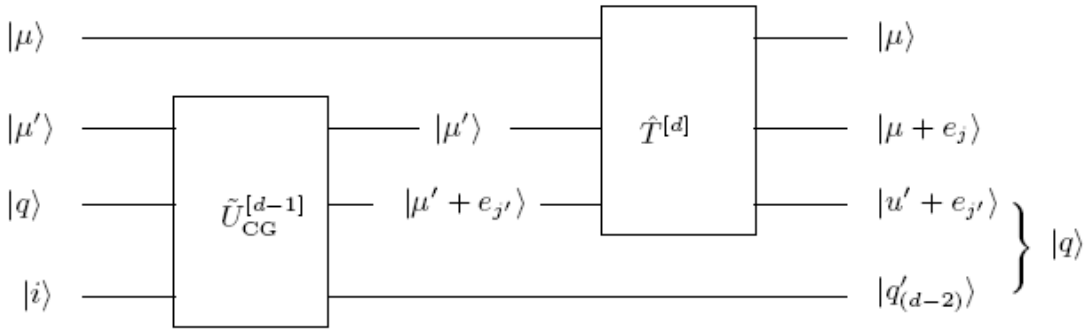



FIG. 4: The $\mathcal{U}_d$ CG transform, $\mathbf{U}_{\mathrm{CG}}^{[d]}$, is decomposed into a $\mathcal{U}_{d-1}$ CG transform $\tilde{U}_{\mathrm{CG}}^{[d-1]}$ (see Eq. (54)) and a reduced Wigner operator $\hat{T}^{[d]}$. In Fig. 5 we show how to reduce the reduced Wigner operator to a $d \times d$ matrix conditioned on $\mu$ and $\mu' + e_{j'}$.

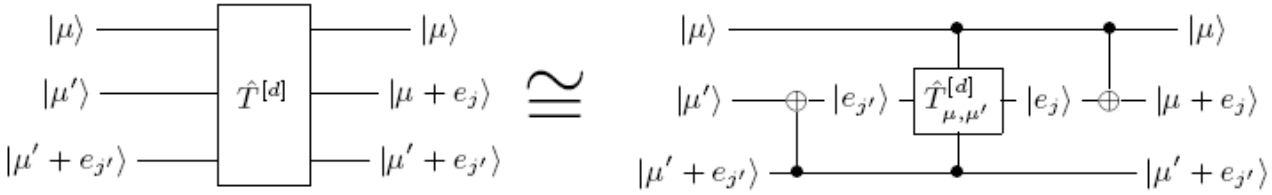



FIG. 5: The reduced Wigner transform $\hat{T}^{[d]}$ can be expressed as a $d \times d$ rotation whose coefficients are controlled by $\mu$ and $\mu' + e_{j'}$.

Schur transform to accuracy $\epsilon$ : time $poly\left(log\frac{1}{\epsilon}, n, d\right)$

# The Plethora of Uses of Schur



Optimal spectrum estimation

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[M. Hayashi and K. Matsumoto. Information Theory, 2004]

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Simulating lattice gauge theories on a quantum computer

[T. Byrnes and Y. Yoshihisa, Phys. Rev. A 2006]

….

**Tr**

**ac**

**ta**

**bl**



# But Algorithms?

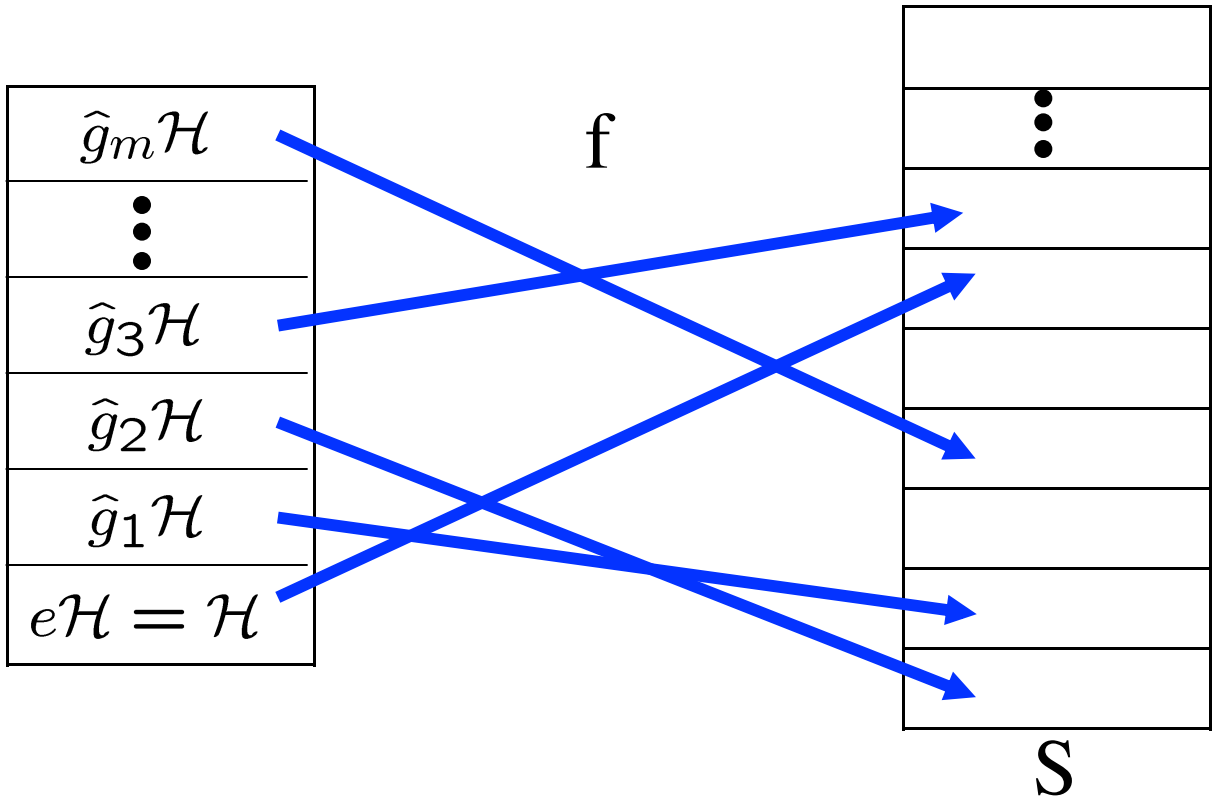
# The Hidden Subgroup Problem

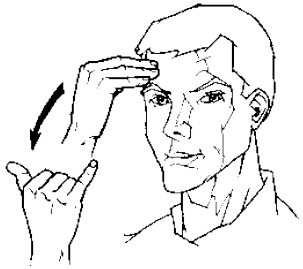


**Setup:** Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$. We say that $f$ hides $\mathcal{H}$.

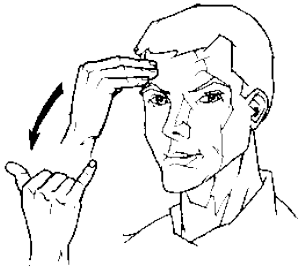
**Problem:** Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").

**Efficient:** If runs in time polylog($|\mathcal{G}|$)





# Why the Fuss?





## Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$ → efficient algorithm for factoring whole numbers

efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ → efficient algorithm for the discrete logarithm problem

Abelian ↑↓ Non-Abelian

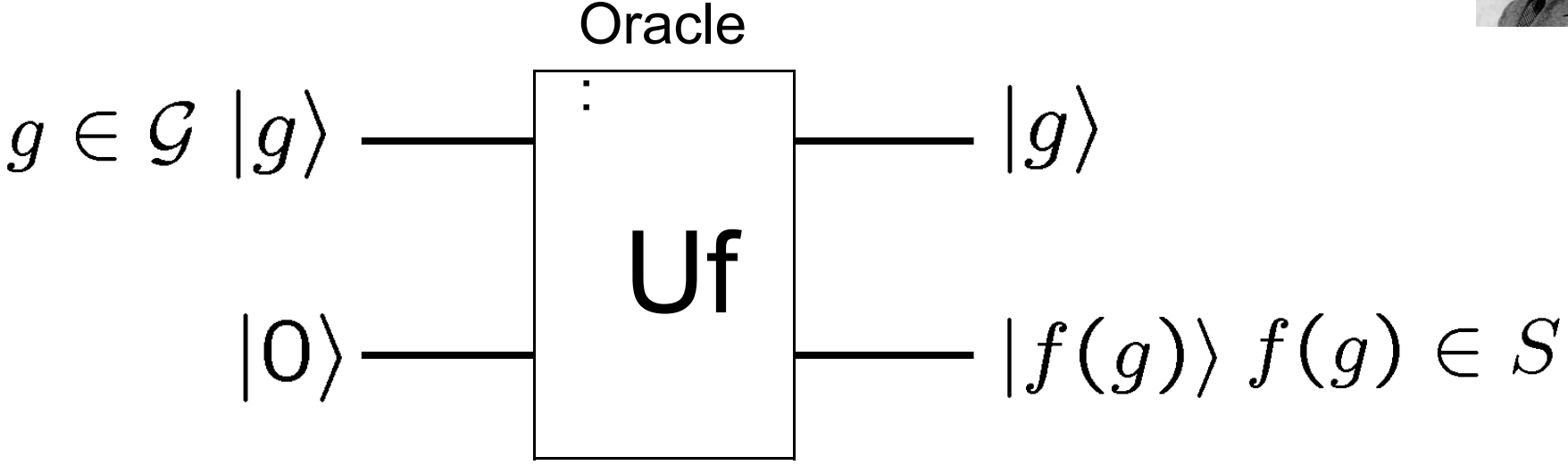
## Open Problems

efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$) → efficient algorithm for the graph isomorphism problem

efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$ → efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

# Standard Query Model






Query in equal superposition over group elements

$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$ hidden subgroup state

# You Down With Symmetry?

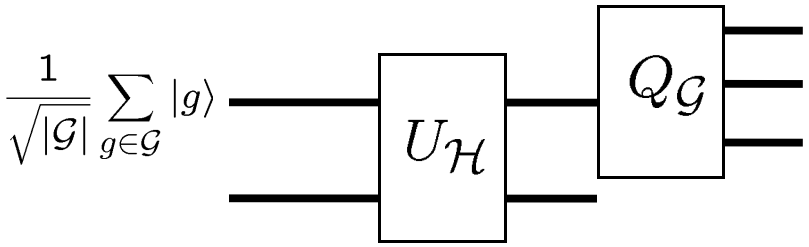


$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \quad \text{hidden subgroup state}$$

Symmetry under left regular representation:

$$D_L(g)|g'\rangle = |gg'\rangle \qquad D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

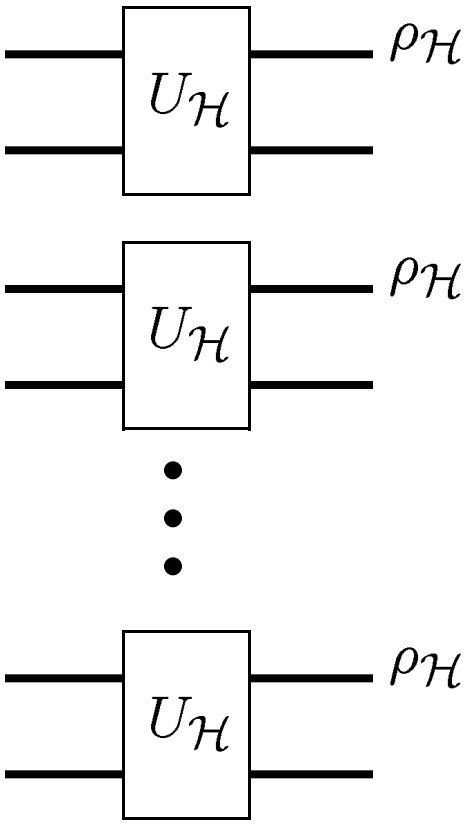
Fourier transform over finite group $\mathcal{G}$




**Can we solve HSP this way?**

# Multicopy





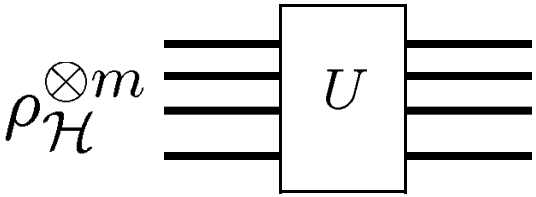

$\rho_{\mathcal{H}}^{\otimes m}$

Can we make a measurement on $m < O(\log |\mathcal{G}|)$ copies and learn the subgroup?

In general (and for important case of symmetric group) **NO**

[Moore, Russell, Hallgren, Roetteler, Sen 05]

**Required:**



# A Tricky HSP

Semidirect product group $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$



Elements of the group:

$$\left( \begin{pmatrix} x \\ y \end{pmatrix}, b \right) \quad x, y, b \in \mathbb{Z}_p$$

Multiplication rule:

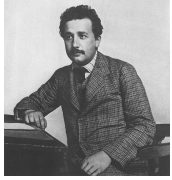
$$\left( \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, b_1 \right) \left( \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_2 \right) =$$

$$\left( \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{b_1} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_1 + b_2 \right)$$

From the play “Copenhagen” Margrethe Bohr to Heisenberg:

“Your talent is for skiing too fast for anyone to see where you are. For always being in more than one position at a time, like one of your particles.”

# Multicopy Success!



Efficient algorithm for the Heisenberg HSP:

[D. Bacon, A.M. Childs, and W. van Dam FOCS 2005]

Optimal measurement approach:

$$\rho_{\mathcal{H}} \qquad Pr(succ) = O\left(\frac{1}{p}\right)$$

$$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}} \qquad Pr(succ) = O(1)$$

Efficient algorithm for the Heisenberg HSP!

**Complicated:** But Why Does It Work?

# Clebsch-Gordan Transform Over Finite Group

Representation:

$$g \to D(g) \qquad D(g_1 g_2) = D(g_1) D(g_2)$$

Irreducible representation (irreps):

$D_\mu(g)$ no invariant subspaces except empty subspace

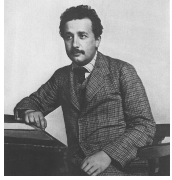
Diagonal product of two irreps:

$$D_{\mu_1}(g) \otimes D_{\mu_2}(g) = \bigoplus I_{n_\mu} \otimes D_\mu(g)$$

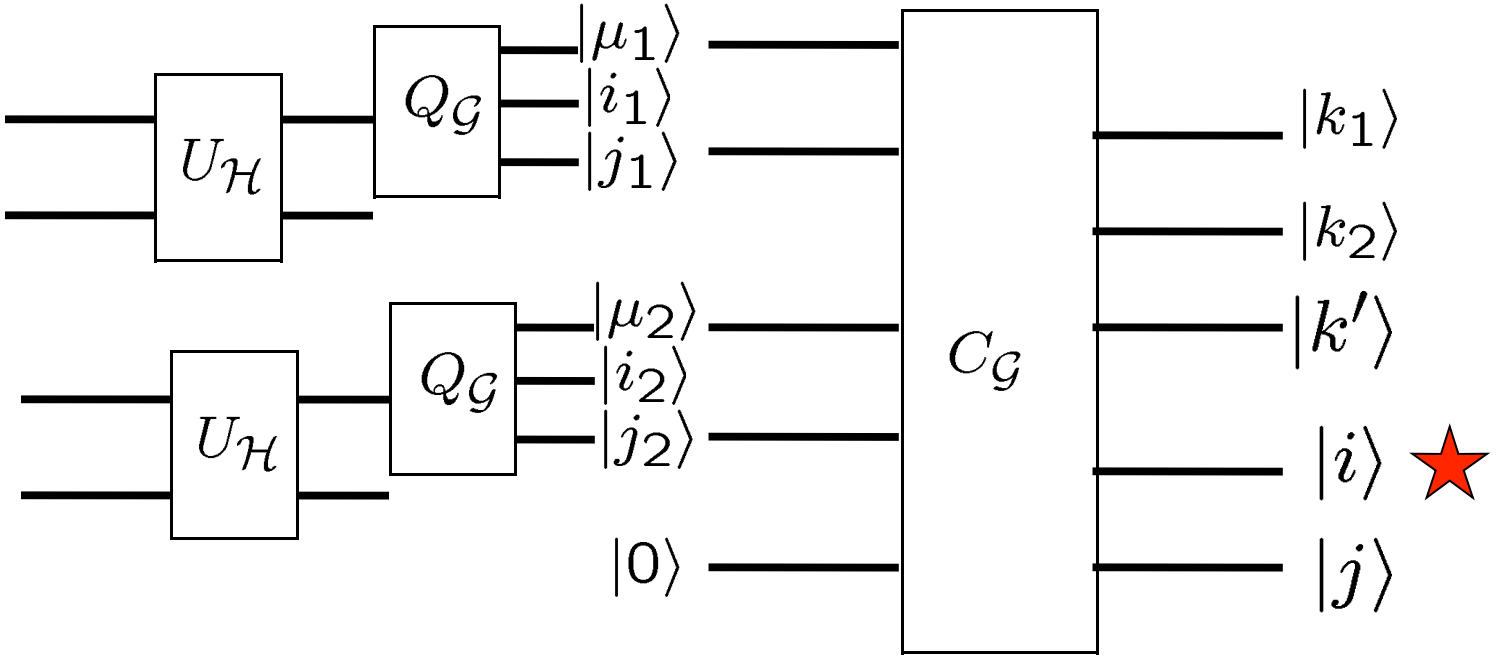
Clebsch-Gordan transform:

$$D_{\mu_1}(g) \otimes D_{\mu_2}(g) = \bigoplus I_{n_\mu} \otimes D_\mu(g)$$

# Entangled Measurement



[Bacon, arXiv:quant-ph/0612107]




Efficient Heisenberg HSP
(Non-Abelian Hidden Subgroup Problem)

# Physicist Are Symmetry Fanatics

Quantum algorithms which offer exponential speedups all exploit symmetries?

What other symmetries can be exploited?

What other techiniques of physics can be exploited? Dream.

# Traditional Excitement

1. Building a quantum computer

2. Quantum computers break cryptosystems

3. Quantum cryptography as physics based security

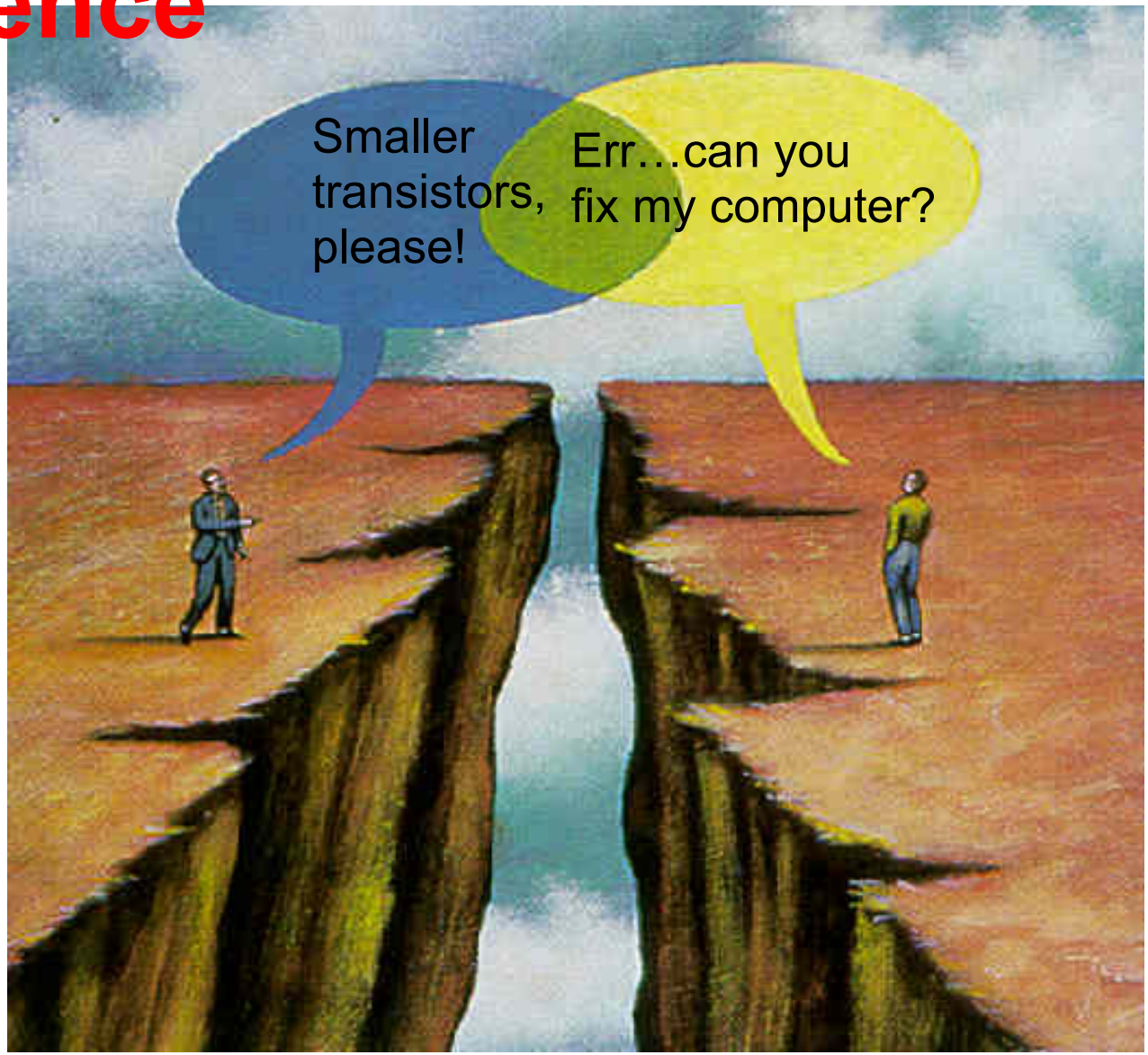
# Emerging Themes

**[REVOLUTION]** Quantum computing as key to progress in theoretical computer science

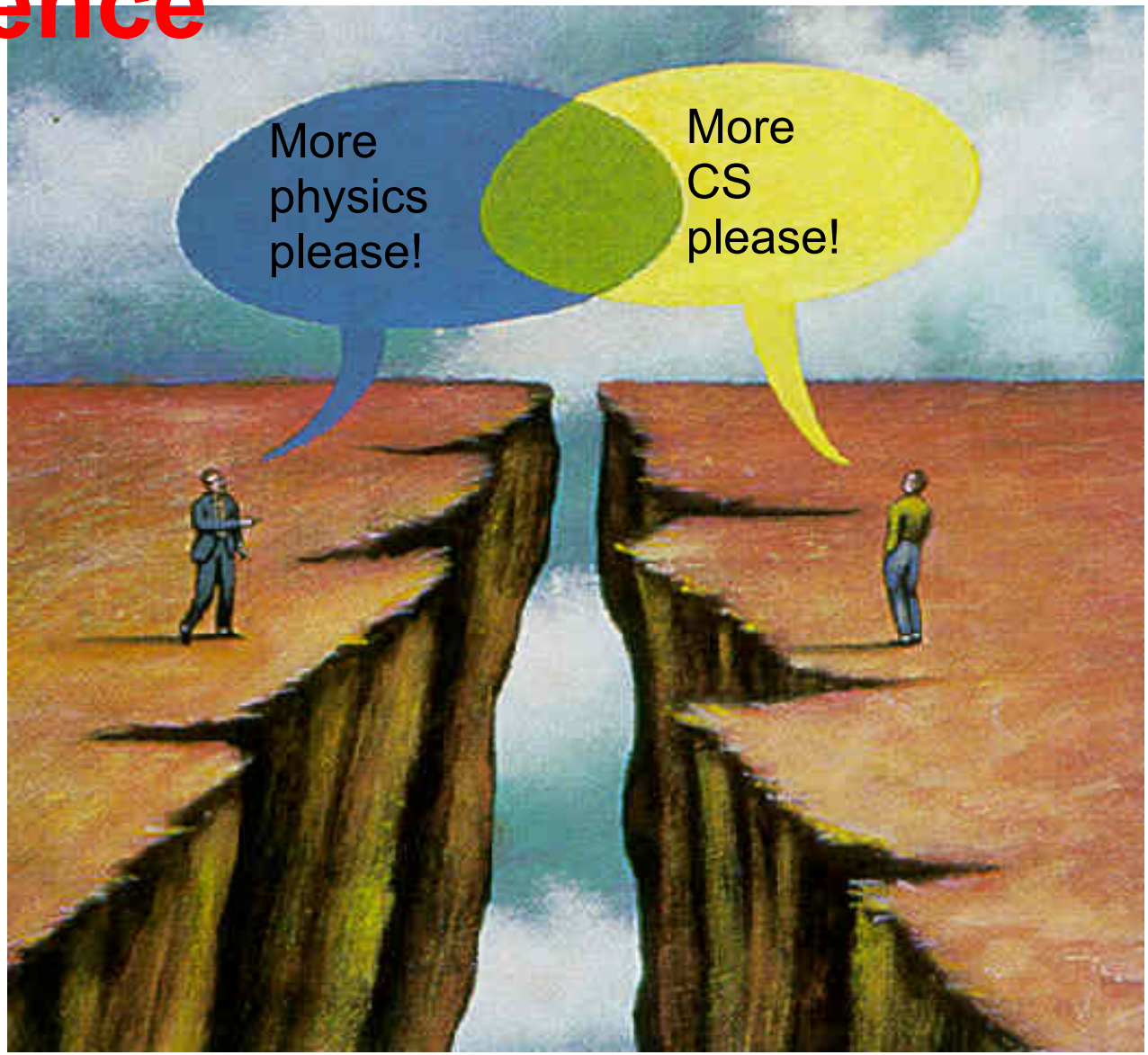
**[STAND ON HEAD]** Understanding power of quantum computers ⇒ new efficient classical algorithms for simulating quantum systems

**[PHYSICISTS!]** Tools and methods developed by physics used for useful computer science (new quantum algorithms)

# Physics and Computer Science

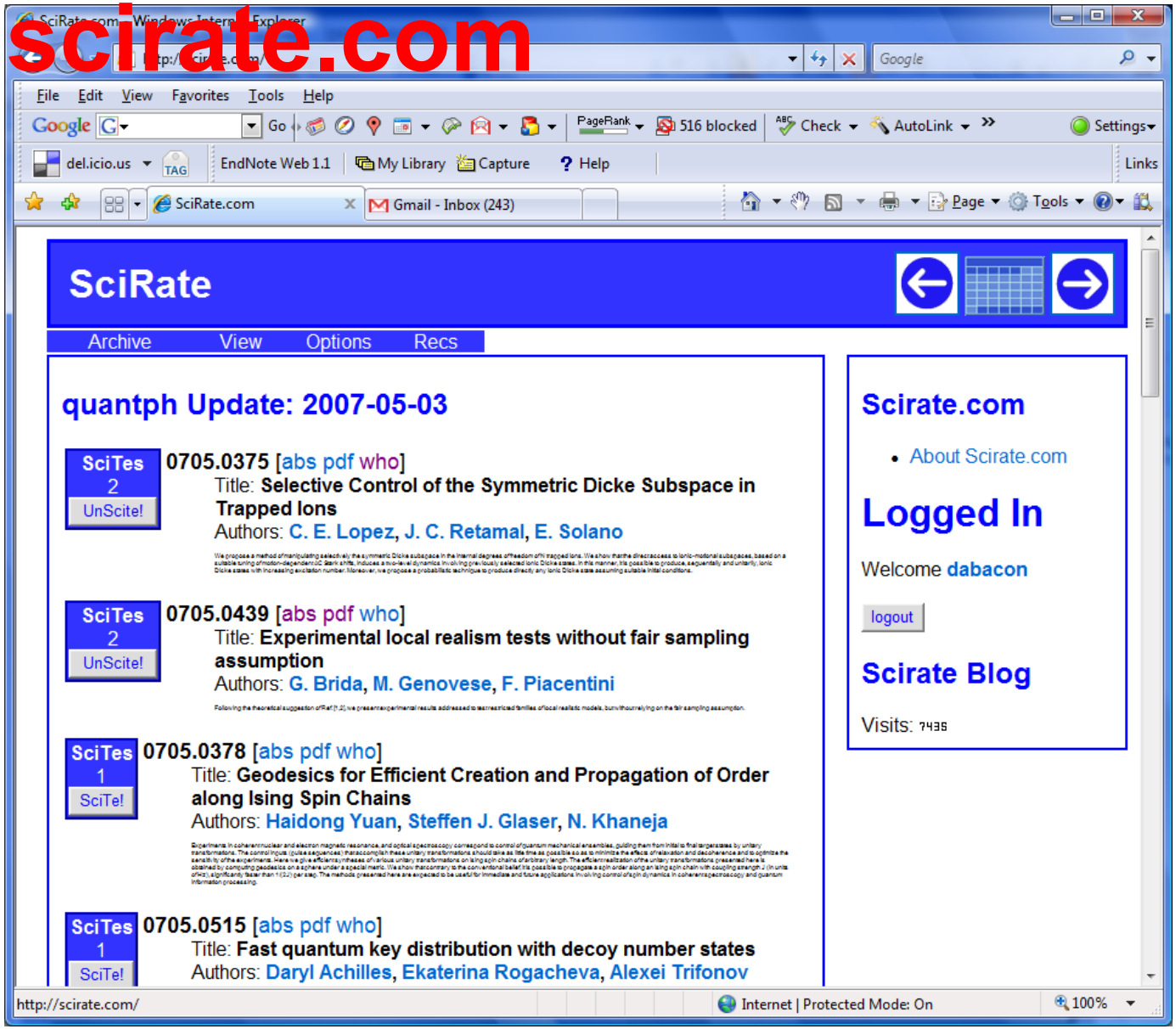


# Physics and Computer Science



# Quick Promo:

## scirate.com



# When Physicists Build Quantum Algorithms

**Dave Bacon**
Department of Computer Science & Engineering
University of Washington

*funded by*





















Collaborators: Childs [Caltech], van Dam [UCSB], Chuang [MIT], Harrow [Bristol]